Multispectral–Hyperspectral Image Fusion via Similarity-Guided Graph Attention and VAE-Transformer

Biwei Chi, Hangyuan Lu, Rixian Liu, Yong Yang, *Senior Member, IEEE*, lingrong Xu, and Weiguo Wan

***Abstract*— Fusion of a high-spatial-resolution multispectral image (MSI) and a low-spatial-resolution hyperspectral image (HSI) aims to generate a high-spatial-resolution HSI (HR-HSI). Most fusion methods use simple upsampling techniques to increase the resolution of HSI without guidance, which can introduce unwanted artifacts and lead to spectral distortion. Additionally, they face challenges in generalization and robustness. To address these challenges, this article introduces a cross-modal fusion network for MSI and HSI, named CSGAV, which is built on similarity-guided graph attention (SGA) and a variational autoencoder-Transformer (VAET). Specifically, we develop a similarity measure algorithm to compute the similarity degree between the source images and construct an SGA module to mitigate modal differences, producing precise upsampled outputs. Moreover, we present an adaptive weighted Transformer, with the weights guided by a variational autoencoder (VAE), thereby enhancing the generalization and robustness of the model. The SGA and VAET are integrated in the cross-modal interactive architecture to achieve the final HR-HSI image. Experimental results conducted on four public datasets show that CSGAV is superior compared to existing state-of-the-art fusion methods both in fusion performance and generalization. The code of this work is available at https://github.com/yotick/CSGAV.**

***Index Terms*— Graph attention, hyperspectral-multispectral image (MSI) fusion, transformer, variational auto encoder.**

I. Introduction

IN REMOTE sensing, hyperspectral image (HSI) and multispectral image (MSI) are powerful tools for capturing detailed information about the Earth's surface [1]. HSI offers fine-grained spectral details across hundreds of contiguous bands, while MSI provides higher spatial resolution with fewer broad spectral bands. It is essential to obtain high spatial resolution HSI (HR-HSI) by fusing high-resolution MSI (HR-MSI) with low-resolution HSI (LR-HSI) [2]. This fusion approach enhances both spatial and spectral resolution, enabling more accurate remote sensing analysis. For instance, in agriculture, it improves crop monitoring and yield prediction, while in environmental monitoring, it allows for better detection of land and vegetation changes. For urban planning, the fusion technique supports detailed land-use mapping, aiding in infrastructure and disaster management [3], [4], [5].

MSI and HSI image fusion (MHIF) methods can be divided into traditional approaches and deep learning-based techniques. Traditional image fusion methods primarily relied on image processing and statistical techniques such as matrix and tensor factorization, spectral unmixing, and so on. For example, Kanatsoulis et al. [6] introduced a coupled tensor decomposition framework that applies canonical polyadic decomposition to model the dimensional dependencies between HSI and MSI, achieving impressive fusion results. Chen et al. [7] presented an MHIF model based on factor-smoothed tensor ring decomposition, which can fully explore the high spatial–spectral correlation of HR-HSI. Xu et al. [8] presented a nonlocal tensor decomposition model for MHIF that improves spatial resolution by sharing factor matrices between the LR-HSI and HR-MSI. Traditional MHIF methods struggle with issues like information loss and degradation estimation, highlighting the need for more effective deep learning approaches [9], [10].

Deep learning approaches, through the construction of complex neural network models, can automatically learn effective feature representations from the data, achieving remarkable results. Wang et al. [11] proposed FusionNet, a variational probabilistic autoencoder framework implemented using convolutional neural networks (CNNs) for MHIF, which can effectively produce the HR-HSI. Furthermore, Nguyen et al. [12] introduce an unsupervised CNN fusion framework that uses Stein's unbiased risk estimation to combine high spatial and spectral resolution images, yielding high-quality results. Tian et al. [13] propose HMPNet, an interpretable model-driven deep network for fusing HSI, MSI, and panchromatic images, which can enhance parameter selection and achieve a balance between spatial and spectral qualities. However, CNN-based techniques are limited by their finite receptive fields, which hinder their ability to model long-range dependencies effectively [14], [15].

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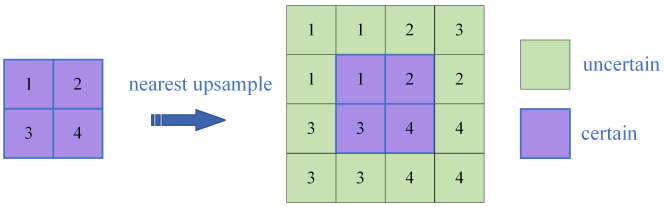


Fig. 1. Process of nearest upsampling.

Recently, researchers are exploring Transformer-based approaches for MHIF. Transformers excel at capturing global interactions and have shown promise in modeling long-range dependencies in image fusion tasks. For instance, Hu et al. [16] propose a Transformer-based method that can globally explore feature relationships by leveraging self-attention mechanisms. Ma et al. [2] propose a novel reciprocal fusion strategy, utilizing directional pairwise multihead cross-attention and Swin Transformer blocks to enhance data integration and self-attention. Deng et al. [17] present the pyramid shuffle-and-reshuffle Transformer (PSRT), which mitigates computational complexity by using fixed-size windows for self-attention and efficient global patch interaction.

In MHIF, to ensure that the spatial sizes of HSI and MSI are consistent, many studies apply the upsampling operation to MSI [18], [19], [20]. However, these traditional upsampling methods, including nearest, bicubic, and bilinear interpolation, often fail to capture the inherent inter-modal relationships between HSI and MSI, limiting the potential for accurate fusion. As shown in Fig. 1, taking nearest upsampling as an example, when performing 2× upsampling, only 1/4 of the points are accurate, while the other points are uncertain and lack any guidance, leading to inaccurate interpolated pixels. Although these algorithms can match the spatial size between MSI and HSI, the interpolated results are imprecise because they rely solely on surrounding pixel information, failing to effectively fuse HSI and MSI. To address this issue, this article designs a novel similarity measure algorithm and proposes a similarity-guided graph attention (SGA) module to fully leverage MSI information during the upsampling process of HSI, thereby further reducing the modal gap between the source images.

Another challenge faced in the MHIF is the generalization of the deep learning-based method [21]. Many methods often struggle to generalize well across different datasets, with poor performance when tested on data that differs from the training set. To mitigate this limitation, this article proposed a variational autoencoder-Transformer (VAET). Unlike the conventional Transformer, we design adaptive weights for the Transformer, which are encoded by a variational autoencoder (VAE). The VAE introduces stochastic noise during the encoding process, which helps the model learn more robust latent representations. This noise forces the model to capture the most essential underlying structures of the data, avoiding overfitting and improving the model's generalization ability. The SGA and VAET modules are integrated in a cross-modal interactive manner to achieve high spatial and spectral fidelity. In summary, the contributions of this article are as follows.

1) We proposed a cross-modal fusion network based on the SGA and VAET, which can effectively integrate relevant information from MSI and HSI with high generalization performance, providing a novel solution to the MHIF problem.
2) We designed the SGA module by developing an SGA architecture, which can mitigate modal differences and effectively achieve more precise upsampling results for HSI.
3) We developed the VAET module by leveraging a Transformer with the adaptive weights encoded by VAE, which can improve the generalization and robustness of the overall model.

II. Related Works

A. Upsampling Method for HSI

Since HSI has high spectral resolution but low spatial resolution, it is necessary to perform spatial upsampling on HSI to match the resolution of MSI. In recent years, many upsampling methods have been proposed. Among them, most use traditional upsampling methods, such as bilinear interpolation, bicubic interpolation, nearest-neighbor interpolation, and so on. In recent years, methods such as MDA-DUNet [22], SSR-NET [23], MSST [24], and MCT-Net [25] have employed bilinear interpolation for upsampling HSI, achieving spatial resolution comparable to HR-MSI. Moreover, bicubic interpolation is widely used for HSI upsampling. For instance, MLKAF-Net [26], UniNet [27], and MoG-DCN [28] all utilize bicubic interpolation for HSI upsampling.

In addition to traditional upsampling methods, deep learning-based methods have been employed recently. For example, Lu et al. [3] applied PixelShuffle in the pansharpening, which can get an enhanced upsampled MSI. Transposed convolutional (TC) [29], [30] is also often employed in image upsampling due to their ability to control the upsampling scale and reduce checkerboard artifacts. Furthermore, several studies have explored implicit neural representations for upsampling and super-resolution. Tang et al. [31] and Wu et al. [32] leveraged implicit neural representations specifically for upsampling tasks. Building upon this foundation, Zhu et al. [33] introduced the quadtree implicit sampling-generative adversarial network (QIS-GAN). Then, Zhu et al. [34] further proposed a guided implicit neural representations method to optimize the upsampling process.

Although the aforementioned methods successfully perform HSI upsampling, they still face challenges, such as inaccurate upsampling information, large parameter sizes, and the need for extra guiding data during the upsampling process. Compared to these methods, the proposed SGA module can mitigate modal differences and effectively achieve more precise upsampling results for HSI, without the need for extra guiding data.

B. Self-Attention in MHIF

In MHIF, Transformers stand out compared to traditional CNNs due to their global feature extraction capabilities and self-attention mechanisms. This allows Transformers to better

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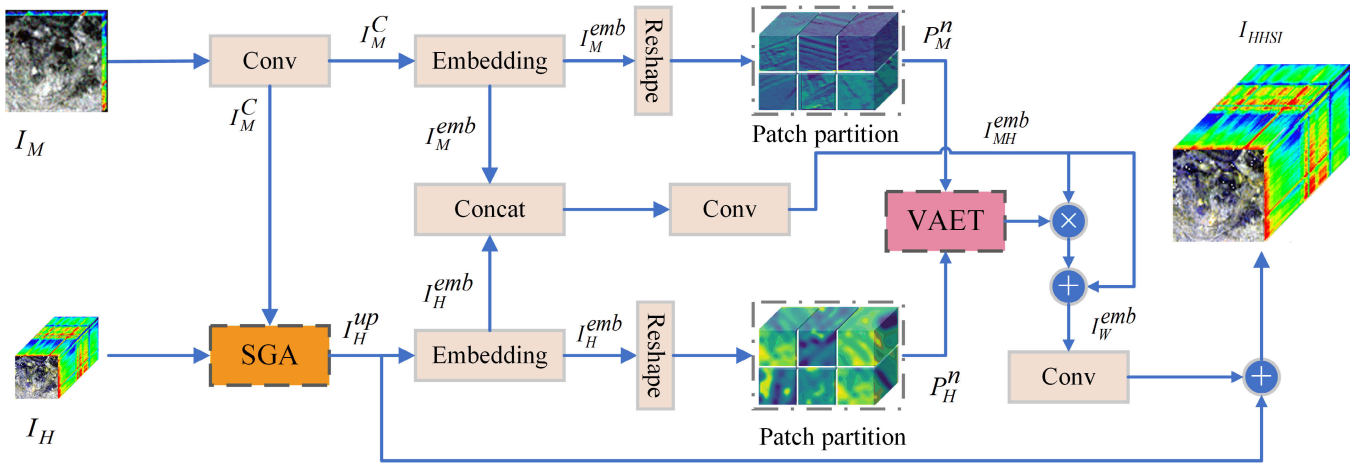


Fig. 2. Overview of the CSGAV architecture. Conv represents a convolution operation, and Concat represents a concatenation operation.

capture long-range dependencies and complex spectral–spatial information in images, thereby enhancing fusion performance and detail preservation. As a result, Transformers have been widely applied in MHIF. Ma et al. [2], Jia et al. [24], Wang et al. [25], and Gao et al. [35] introduced different Transformer-based networks for MHIF, highlighting the importance of incorporating Transformer architectures in this domain. Moreover, Fang et al. [19] presented the concept of a multiinput and multioutput spatial–spectral Transformer to capture details and structures effectively in images. Cao et al. [20] proposed an unsupervised hybrid model that integrates a Transformer and CNN for unsupervised MHIF, emphasizing the multilevel spatial–spectral relationships between HR-HSI and the observed images. Sun et al. [36] proposed a multiscale deep cross-fusion transformer network for HSI and MSI fusion. This network effectively performs the interactive fusion of spatial–spectral features, thereby enhancing the quality of the fused images. Lately, Sun et al. [37] introduced an information-aware Transformer-based unfolding network, focusing on modeling long-range dependencies and transferring more information across stages. Li et al. [38] further proposed a multiscale hybrid network that optimized information processing through a spectral splitting mechanism. In conclusion, the integration of Transformer architectures in MHIF has shown promising results in enhancing image resolution and quality, demonstrating the potential for further advancements in this field.

III. Methodology

In this section, we will introduce the main blocks and mechanisms of the proposed method, together with the adopted loss function.

A. Overview

For convenience, we denote the input MSI as $I_M \in \mathbb{R}^{c\times H\times W}$, where c, H, and W denote the spectral bands, the numbers of rows, and columns, and the input HSI as $I_H \in \mathbb{R}^{C\times h\times w}$, where C, h, and w are the spectral bands, the number of rows, and columns. The flowchart of CSGAV is shown in Fig. 2.

This network first expands I_M in the channel dimension using a convolution operation, obtaining $I_M^C \in \mathbb{R}^{C\times H\times W}$. Then, I_M^C and I_H are fed into the SGA module to get a more accurate upsampling result $I_H^{\text{up}} \in \mathbb{R}^{C\times H\times W}$.

Next, I_M^C and I_H^{up} are passed through the embedding layers to generate $I_M^{\text{emb}} \in \mathbb{R}^{E\times H\times W}$ and $I_H^{\text{emb}} \in \mathbb{R}^{E\times H\times W}$, respectively. Then, I_M^{emb} and I_H^{emb} are resized as $I_M^{\text{emb}}, I_H^{\text{emb}} \in \mathbb{R}^{E\times(HW)}$, which can be conveniently divided into several patches, denoted as $P_M^n \in \mathbb{R}^{E\times(HW/N)}$ and $P_H^n \in \mathbb{R}^{E\times(HW/N)}$, respectively, where n is the order number of individual patches and N represents the total number of patches.

To fuse the information of the two modalities, I_M^{emb} and I_H^{emb} are concatenated and through a convolutional layer to get $I_{\text{MH}}^{\text{emb}} \in \mathbb{R}^{E\times H\times W}$. Subsequently, the VAET module is used for fusion of P_M^n and P_H^n, resulting in a combined weight. This weight is first multiplied by $I_{\text{MH}}^{\text{emb}}$, and the resulting product is then added to $I_{\text{MH}}^{\text{emb}}$ to obtain a cross-modal weighted results $I_W^{\text{emb}} \in \mathbb{R}^{E\times H\times W}$.

Finally, I_W^{emb} is passed through a convolution operation, then added to I_H^{up} and reconstructed to obtain the final high-resolution HR-HSI $I_{\text{HHSI}} \in \mathbb{R}^{C\times H\times W}$.

B. SGA Module

To achieve more refined upsampling results for HSI and reduce the modality difference, we propose the SGA module, as illustrated in Fig. 3. Unlike the existing methods, the SGA module incorporates a similarity-guided mechanism based on the graph attention convolution network [39]. This mechanism allows the SGA to dynamically adjust the attention weights based on the similarity between nodes, ensuring that the most relevant spatial and spectral features from the MSI data are effectively utilized to guide the HSI upsampling process. By focusing on these key features, the SGA module helps to reduce modality differences between the HSI and MSI datasets and facilitates the fusion of their crucial information. The specific process is as follows.

First, the input I_H is upsampled to four times its original size and then reshaped to obtain $I_H^R \in \mathbb{R}^{C\times(HW)}$; meanwhile, I_M^C was also reshaped to obtain $I_M^R \in \mathbb{R}^{C\times(HW)}$. To measure and leverage cross-modal feature relationships, Gram matrices

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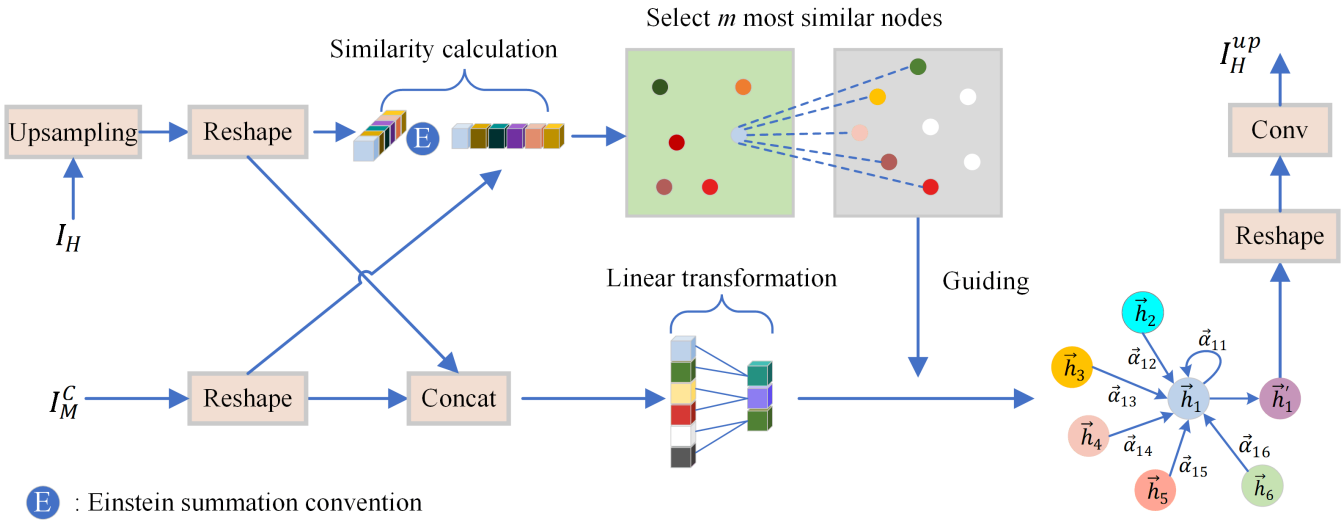


Fig. 3. Flowchart of the SGA module.

[40] are introduced to analyze second-order correlations and characterize statistical structures within modalities. For I_H^R and I_M^R, their respective Gram matrices are defined as

$$G_H = I_H^R \left(I_H^R\right)^T \in \mathbb{R}^{(HW)\times(HW)} \tag{1}$$

$$G_M = I_M^R \left(I_M^R\right)^T \in \mathbb{R}^{(HW)\times(HW)}. \tag{2}$$

Then, to establish cross-modal spatial–spectral correspondences, we develop a similarity metric algorithm using the Einstein summation convention, constructing a similarity matrix $S = I_H^R (I_M^R)^T$ that simultaneously functions as a cross-modal Gram matrix. The matrix S encodes pairwise correlations through its elements

$$S_{ij} = \sum_{k=1}^{C} I_H^R(i,k) \cdot I_M^R(j,k) \tag{3}$$

where i, j index the spatial positions and k denotes the spectral band dimension. This formulation intentionally leverages HSI spectral information to guide graph structure construction for adaptive feature fusion. To ensure that the graph structure effectively represents the most pertinent features, this module selects the m most similar MSI nodes for each HSI node to guide the construction of the graph structure. Next, I_H^R and I_M^R are concatenated, followed by a linear transformation layer to obtain the fused features. These nodes are connected based on the graph structure constructed before, then the edge weights e_{ij} are computed as follows:

$$e_{ij} = \sigma\left(a^T \left[W\vec{h}_i \,||\, W\vec{h}_j\right]\right) \tag{4}$$

where $\vec{h}_i$ and $\vec{h}_j$ represent the features of nodes i and j, respectively, W is a learnable weight matrix, and a is a learnable attention vector. $||$ denotes the concatenation operation, and $\sigma(\cdot)$ is a nonlinear activation function used to introduce nonlinearity.

Furthermore, to focus on the most significant connections, the computed edge weights e_{ij} are normalized to obtain the attention weights $\vec{\alpha}_{ij}$. Then, the weighted node $\vec{h}_i' \in \mathbb{R}^{C\times(HW)}$ is calculated by weighted summation of the features of its neighboring nodes

$$\begin{cases} \vec{\alpha}_{ij} = \text{softmax}(e_{ij}) \\ \vec{h}_i' = \sigma\left(\sum_{j\in N(i)} \vec{\alpha}_{ij} W \vec{h}_j\right) \end{cases} \tag{5}$$

where softmax$(\cdot)$ represents the softmax function, and $N(i)$ represents the set of neighbors of node i.

The relationship between $\vec{h}_i'$ and $\vec{h}_i$ can be expressed as $\vec{h}_i' = \Delta\vec{h} + \vec{h}_i$, where $\Delta\vec{h}$ represents the feature increment of node i. As a result, the collection of $\vec{h}_i'$, called $I_H^{R'}$ can be updated by

$$I_H^{R'} = I_H^R + \Delta I_H^R \tag{6}$$

where ΔI_H^R is a matrix composed of the feature increments $\Delta\vec{h}$ of each node, representing the increment part of the feature matrix I_H^R after graph convolution. By dynamically updating the feature matrix I_H^R through graph convolution, the corresponding Gram matrix $G_H = I_H^R (I_H^R)^T$ is updated to $G_H' = I_H^{R'} (I_H^{R'})^T$. This transformation effectively reduces the discrepancy between the HSI and MSI Gram matrices, making G_H' approximate G_M gradually. As a result, it enables accurate cross-modal feature alignment while preserving spectral–spatial correlations.

After the graph convolution operation, the module reshapes the fused features $\vec{h}_i'$ to the size of $\mathbb{R}^{C\times H\times W}$ and get the final output I_H^{up} through a convolution operation.

C. VAET Module

To address the issues of limited generalization and robustness in MHIF, this section presents the VAET module. Specifically, VAET can be divided into two stages, and the overall architecture of VAET can be found in Fig. 4. First, P_M^n and P_H^n are fed into the linear layers of the Transformer module to obtain their query vectors (Q), key vectors (K), and

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value vectors (V) for each

$$Q_1, K_1, V_1 = \text{Linear}\left(P_M^n\right) \tag{7}$$

$$Q_2, K_2, V_2 = \text{Linear}\left(P_H^n\right). \tag{8}$$

Subsequently, regulated randomness noise is introduced to enhance the network's robustness against data variability. This noise serves as a regularization technique, preventing the model from becoming overly reliant on specific features or patterns in the training data, which encourages more robust and generalized representations [41]. This is accomplished by passing P_M^n and P_H^n through the encoder part of VAET to derive three sets of weights corresponding, respectively, to Q, K and V. The detailed steps of this approach can be seen as follows: first, P_M^n is passed through the neural network encoder (NN Encoder), which consists of linear layers, to produce the encoded mean values μ_1, μ_2, μ_3 and the corresponding variances σ_1, σ_2, and σ_3. The same operation can also be performed on P_H^n to obtain $\sigma_4, \sigma_5, \sigma_6, \mu_4, \mu_5$, and μ_6.

Then, to introduce variability, noise e_i for $i = 1, \ldots, 6$ is sampled from a standard normal distribution, i.e., $e_i \sim \mathcal{N}(0, 1)$ for $i = 1, \ldots, 6$. These noises are subsequently multiplied by the variances σ_i to generate perturbation terms, given by $\sigma_i \cdot e_i$ for $i = 1, \ldots, 6$. The original encodings are then added to these perturbation terms to obtain the resulting weights, represented as

$$W_i = \mu_i + \sigma_i \cdot e_i. \tag{9}$$

To further regularize the latent representation, we introduce the Kullback–Leibler (KL) divergence as a penalty term. The KL divergence measures the difference between the learned latent distribution and a standard normal distribution. Specifically, for each latent variable, the KL divergence is computed as

$$\mathcal{L}_{\text{KL}} = \sum_{i=1}^{6} \frac{1}{2}\left(\mu_i^2 + \sigma_i^2 - \log\left(\sigma_i^2\right) - 1\right). \tag{10}$$

This regularization term encourages the model's latent representation to align with the standard normal distribution, thereby structuring the latent space more effectively and improving generalization. By constraining the latent distribution, the KL divergence enhances the model's generative capacity and stability, reducing overfitting [42], [43].

To better fuse cross-modal information, the module integrates perturbation-enhanced features through cross-weighting. Specifically, Q_1, K_1, and V_1 are each multiplied by W_4, W_5, and W_6, respectively, to produce Q_1', K_1', and V_1'. Similarly, Q_2, K_2, and V_2 are each multiplied by W_1, W_2, and W_3, respectively, resulting in Q_2', K_2', and V_2'. These calculations can be expressed mathematically as follows:

$$Q_1' = Q_1 \cdot W_4, \quad K_1' = K_1 \cdot W_5, \quad V_1' = V_1 \cdot W_6 \tag{11}$$

$$Q_2' = Q_2 \cdot W_1, \quad K_2' = K_2 \cdot W_2, \quad V_2' = V_2 \cdot W_3. \tag{12}$$

The attention weights for P_M^n and P_H^n, denoted as $P_{M\text{-att}}^n$ and $P_{H\text{-att}}^n$, can be obtained by

$$P_{M\text{-att}}^n = \text{softmax}\left(\frac{Q_1' \cdot K_1'}{\sqrt{d_k}}\right) \cdot V_1' \tag{13}$$

$$P_{H\text{-att}}^n = \text{softmax}\left(\frac{Q_2' \cdot K_2'}{\sqrt{d_k}}\right) \cdot V_2' \tag{14}$$

where d_k represents the dimensionality of the key vectors.

Finally, the final attention weights can be obtained by combining $P_{M\text{-att}}^n$ and $P_{H\text{-att}}^n$, as follows:

$$P_{\text{All-att}}^n = P_{M\text{-att}}^n + P_{H\text{-att}}^n. \tag{15}$$

D. Loss Function

This article uses a composite loss function to train the network, which combines an L_1 reconstruction loss and a KL divergence regularization term. Specifically, the loss function is defined as

$$L = L_1\text{loss}\left(I_{\text{HHSI}}, I'_{\text{HHSI}}\right) + \alpha \cdot \mathcal{L}_{\text{KL}} \tag{16}$$

where I_{HHSI} denotes the reconstructed HSI generated by the model, and I'_{HHSI} is the corresponding ground truth. The coefficient α is a dynamic weighting parameter that adjusts the contribution of the KL divergence loss $\mathcal{L}_{\text{KL}}$ during training.

IV. Experimental Results

This section is mainly dedicated to presenting the experimental settings and results in our experiments. First, experiments were conducted on three hyperspectral simulation datasets and one real-world dataset, with the results compared against those of ten fusion methods. Additionally, ablation experiments were conducted on each module within the network, and the impact of different modules on the effect of the network was further explored. Finally, the model's generalization performance was validated across different datasets.

A. Experimental Settings

1) The network is developed using PyTorch 1.11.0 and Python 3.8, optimized with AdamW at a 0.0001 learning rate, and trained over 150 epochs. The system operates on Windows with an NVIDIA GeForce RTX 3090 GPU. To ensure the fairness of the comparisons, all the parameters of the comparison methods were kept consistent with their original implementations.
2) *Datasets:* To demonstrate the effectiveness of the proposed method, we assess the performance on three remote sensing HSI datasets: Botswana, Houston, Pavia Center, and a real remote sensing dataset, WorldView-3. The Botswana dataset, captured by the Hyperion sensor on the EO-1 platform, features a 30-m pixel resolution. Out of the 242 bands acquired by the sensor, this study utilized 145 bands following the elimination of uncalibrated and noisy bands. The Houston dataset, used for HSI classification, contains imagery with 144 spectral bands spanning wavelengths from 380 to 1050 nm, calibrated to at-sensor spectral radiance units. The Pavia Center dataset, captured by the ROSIS sensor during a flight over northern Italy, consists of 102 spectral bands with a spatial resolution of 1.3 m per pixel and covers an area of 1096 × 1096 pixels. The WorldView-3 dataset contains eight bands, with a pixel resolution of 0.31 m for panchromatic images, and 1.24 m for MSIs.

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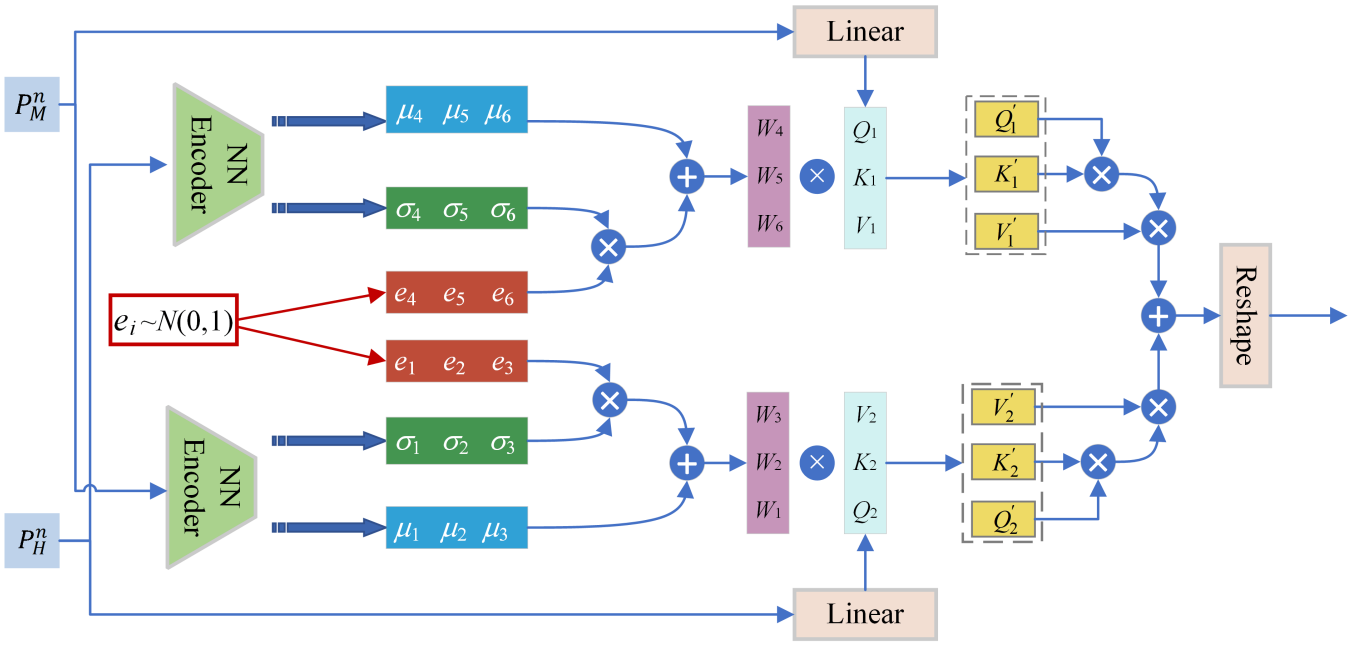


Fig. 4. Flowchart of the VAET module.

3) *Data simulation:* In the Botswana dataset, we selected several 64 × 64 × 145 patches from the original dataset by cropping, rotating, and mirroring operations to create reference HR-HSI images. Then, to produce LR-HSI images, following the method described in [2], we first apply an 8 × 8 Gaussian anti-aliasing filter with a standard deviation of 2 to the HR-HSI images, and then downsample the images by a factor of 4 along both the horizontal and vertical directions. Finally, we select the first four dimensions of the HR-HSI images' channel dimension as the HR-MSI images. Thus, the HR-HSIs, LR-HSIs, and HR-MSIs are of sizes 64 × 64 × 145, 16 × 16 × 145, and 64 × 64 × 4, respectively. Similarly, we use the same operation on the Houston and Pavia Center dataset. In the Houston dataset, the sizes of HR-HSIs, LR-HSIs, and HR-MSIs are 64 × 64 × 144, 16 × 16 × 144, and 64 × 64 × 4, respectively. In the Pavia Center dataset, the sizes of HR-HSIs, LR-HSIs, and HR-MSIs are 64 × 64 × 102, 16 × 16 × 102, and 64 × 64 × 4, respectively. In each dataset, we select 70% of the image pairs for training and 30% for testing.

B. Evaluation Metrics

To quantitatively evaluate the proposed method, this article introduces four metrics: peak signal-to-noise ratio (PSNR↑), universal image quality index (UIQI↑), spectral angle mapper (SAM↓), and Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS↓) [3], [44]. The upward arrow (↑) signifies that higher values correspond to better fusion performance, while the downward arrow (↓) indicates the opposite. To this end, we define $F_k(i, j)$ to denote the reference HSI at the k-band and position (i, j), while $\hat{F}_k(i, j)$ is used to denote the estimated HSI at the same k-band. The specific introduction to these metrics is as follows.

1) PSNR measures the proportion of peak power to the noise power in the fused image, providing insights into the level of distortion occurring during the fusion process. Higher values of PSNR indicate less amounts of distortion, and the fusion results have better spatial detail information. The PSNR of kth band is defined as

$$\text{PSNR} = 10\log_{10}\left(\frac{\max(F_k)^2}{\frac{1}{HW}\|F_k - \hat{F}_k\|_2^2}\right) \tag{17}$$

where H and W represent the height and width of the image, respectively. $\|\cdot\|_2$ is the $L2$-norm.

2) UIQI calculates the similarity between two images based on their contrast, brightness, and structural information. The value ranges from 0 to 1, with higher values indicating more similar images. The calculation formula for UIQI is as follows:

$$\text{UIQI} = \frac{4\sigma_1\sigma_2 C}{(\sigma_1^2 + \sigma_2^2)(C^2 + K)} \tag{18}$$

where σ_1 and σ_2 represent the standard deviations of the two images, C is the covariance, and K is a constant, usually taken as 0.01.

3) ERGAS is a comprehensive metric to assess the average error and dynamic range variation between the fused result and the reference HR-HSI. A lower value of the ERGAS metric signifies a higher degree of similarity between the fused image and the reference image, and it can be defined as

$$\text{ERGAS} = 100 \cdot h/l \cdot \sqrt{\frac{1}{B}\sum_{k=1}^{B}\left(\frac{\text{RMSE}(F_k, \hat{F}_k)}{\mu_k}\right)^2} \tag{19}$$

where h/l is the ratio of the resolution of MSI and HSI, μ_k denotes the mean value of the kth band of HSI, and

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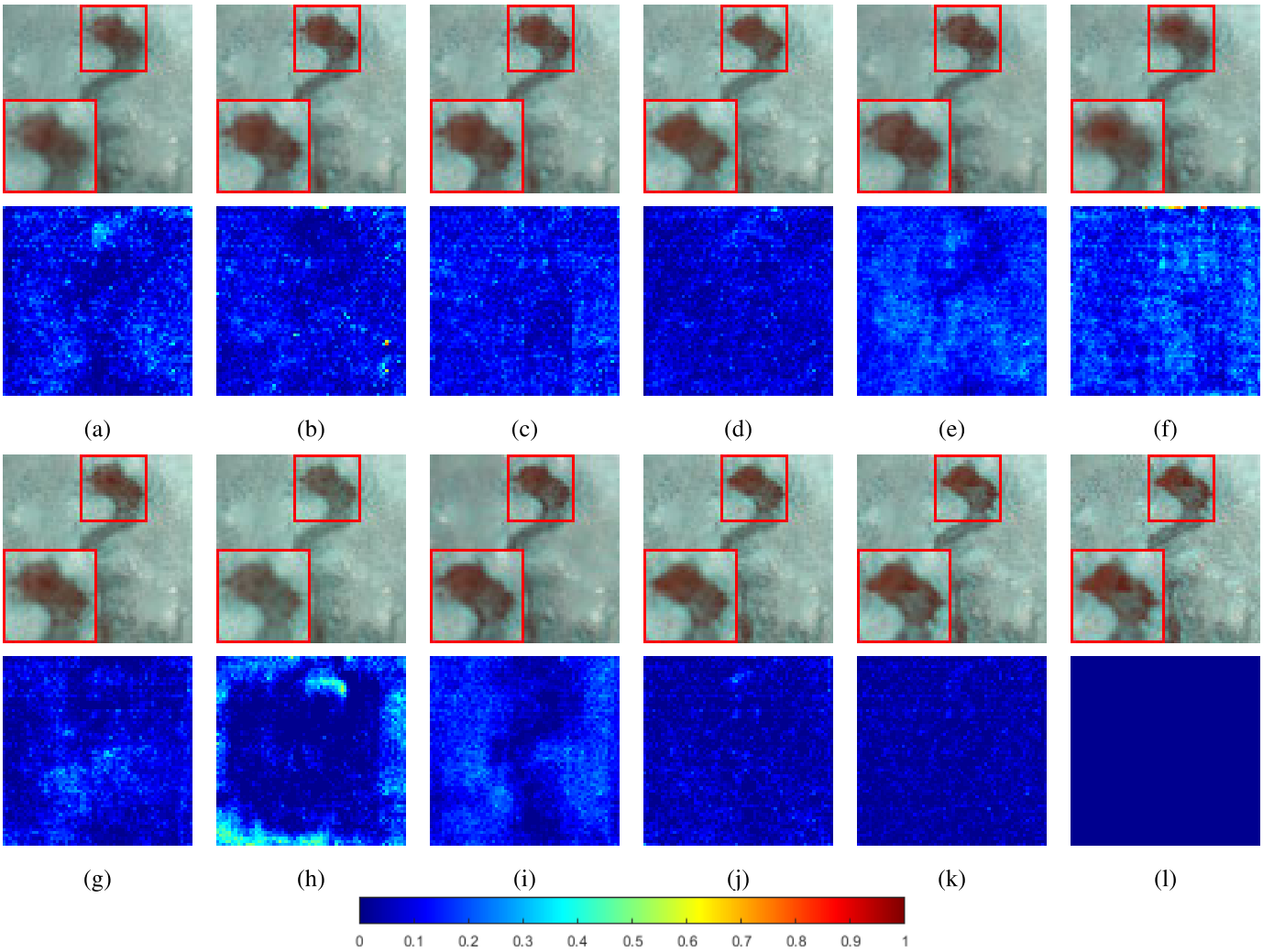


Fig. 5. Fusion results of different fusion methods on the Botswana dataset. The first row displays the pseudo-color RGB images of the fusion results, while the second row presents the corresponding absolute error maps. (a) SSR-NET, (b) Fusformer, (c) PSRT, (d) MCT-Net, (e) QIS-GAN, (f) CSSNet, (g) AMGSGAN, (h) SDAGE, (i) NGST_Net, (j) CYformer, (k) proposed, and (l) GT.

B denotes the number of bands, $\mathrm{RMSE}(F_k, \hat{F}_k)$ denotes the root mean square error between the kth band of the fused image F_k and the reference image $\hat{F}_k$.

4) SAM assesses the spectral accuracy of the fusion result by measuring the angle between the reference and fused images. A lower value of SAM indicates successful reconstruction of spectral information during the fusion process. The formula for computing SAM is presented below

$$\mathrm{SAM} = \arccos\left(\frac{\langle F(i,j), \hat{F}(i,j)\rangle}{\|F(i,j)\|_2 \|\hat{F}(i,j)\|_2}\right) \tag{20}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product operation. The final SAM results are obtained by averaging the sum of SAM values for each pixel.

C. Experimental Results on Simulated Datasets

To assess the performance of our approach, we compare it with various state-of-the-art (SOTA) methods for MHIF. The comparison includes ten methods: SSR-NET [23], Fusformer [16], PSRT [17], MCT-Net [25], QIS-GAN [33], CSSNet [45], AMGSGAN [46], SDAGE [47], NGST_Net [48], and CYformer [49]. Tables I–III present the evaluation metrics of various fusion methods across different datasets, while Figs. 5–7 display the visual fusion outcomes. To enable a thorough comparison, we provide the pseudo-color RGB image of the fusion result along with the corresponding absolute error maps between the fusion result and the reference image.

1) Results on Bostwana Dataset: This section analyzes the visual results on the Botswana dataset, as shown in Fig. 5. Table I provides the quantitative evaluation metrics. The red rectangles represent enlarged close-ups of the highlighted regions to better visualize spatial and spectral fidelity. From Fig. 5, the results of SSR-NET, Fusformer, QIS-GAN, CSSNet, and SDAGE exhibit noticeable spatial distortion, particularly in the highlighted regions, where the overall color is biased. These methods struggle to preserve sufficient spatial details, leading to blurry outputs. PSRT, NGST_Net, and AMGSGAN demonstrate slight improvements in spatial structures. MCT-Net and CYformer offer better spectral–spatial fidelity but there are still observable distortions. Our proposed

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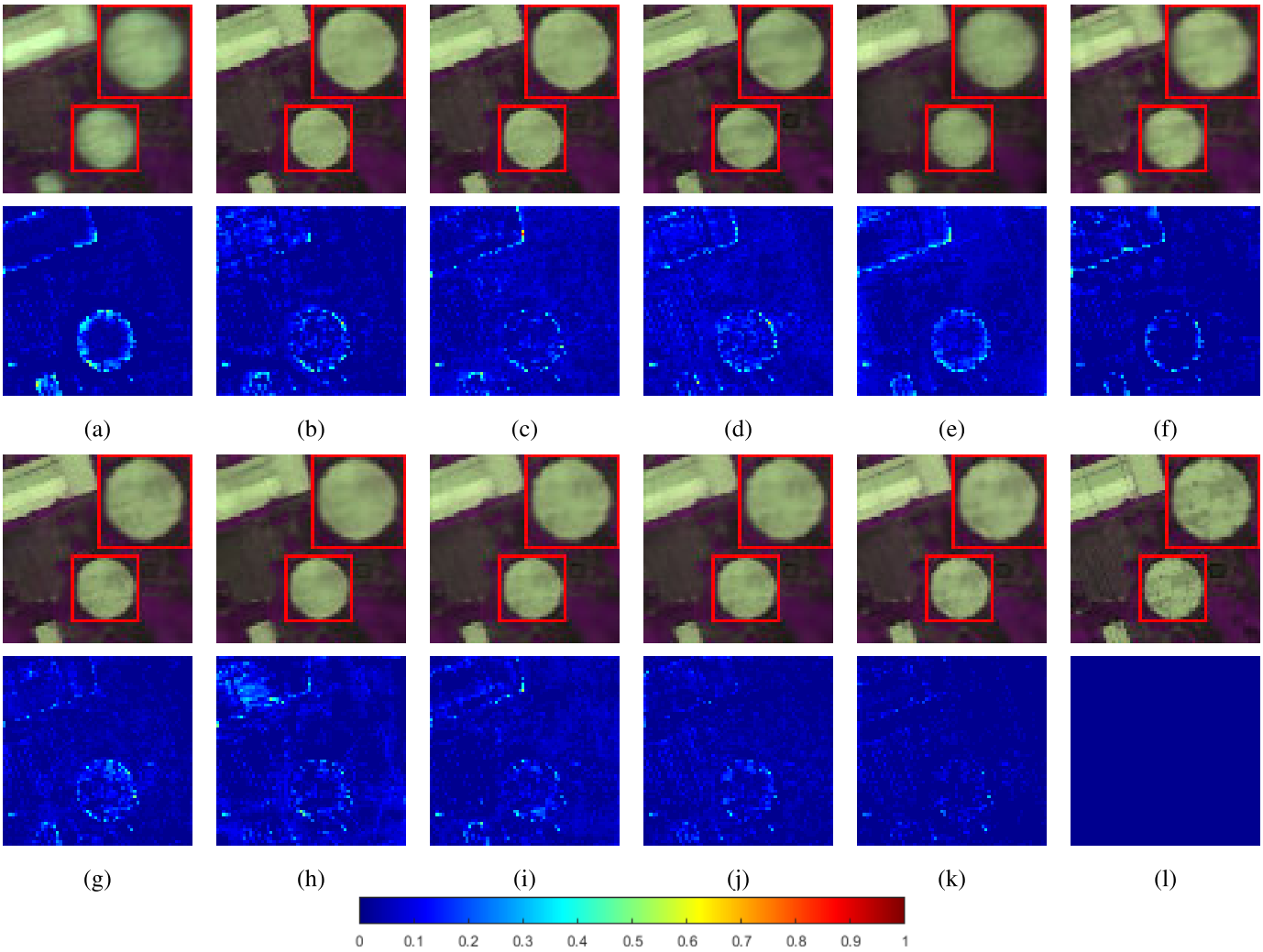


Fig. 6. Fusion results of different fusion methods on the Houston dataset. The first row displays the pseudo-color RGB images of the fusion results, while the second row presents the corresponding absolute error maps. (a) SSR-NET, (b) Fusformer, (c) PSRT, (d) MCT-Net, (e) QIS-GAN, (f) CSSNet, (g) AMGSGAN, (h) SDAGE, (i) NGST_Net, (j) CYformer, (k) proposed, and (l) GT.

TABLE I

EXPERIMENTAL RESULTS ON THE BOTSWANA DATASET, WITH THE BEST VALUES MARKED IN BOLD

Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓	Paras(M)
SSR-NET(2021)	33.452	0.964	2.546	0.876	0.568
Fusformer(2022)	36.080	0.965	1.895	0.603	0.166
PSRT(2023)	36.915	0.965	1.739	0.532	0.314
MCT-Net(2023)	35.843	0.952	1.795	0.666	6.993
QIS-GAN(2023)	34.405	0.962	1.844	0.762	0.380
CSSNet(2023)	37.696	0.972	1.801	0.492	2.275
AMGSGAN(2023)	37.564	0.975	1.827	0.528	4.068
SDAGE(2024)	33.597	0.938	2.206	0.791	6.760
NGST_Net(2024)	36.792	0.969	1.711	0.568	0.222
CYformer(2025)	40.189	0.984	1.731	0.408	0.874
Proposed	**42.554**	**0.986**	**1.600**	**0.282**	2.134

method demonstrates a significantly better balance between spectral and spatial information, closely matching the GT with minimal visual differences. Besides, compared with the second-best performing method, CYformer, the PSNR↑ metric improved by 5.9%, the ERGAS↓ metric decreased by 30.9%, and the SAM↓ metric decreased by 7.6%, indicating a significant enhancement in performance. This clearly demonstrates that CSGAV achieves satisfactory visual fusion results.

2) Results on Houston Dataset: Fig. 6 presents the fusion results and their corresponding absolute error maps. Table II presents the objective evaluation metrics and the parameters for the fusion results of the Houston dataset. It can be observed that SSR-NET, PSRT, QIS-GAN, and CSSNet exhibit clear signs of spatial distortion, particularly in the highlighted regions. In contrast, Fusformer, SDAGE, NGST-Net, and AMGSGAN offer improved spatial preservation, but still show slight spatial distortions. MCT-Net and CYformer provide improved spectral–spatial fidelity, but it struggles to capture finer texture details effectively. In contrast, our proposed method exhibits superior spatial accuracy and produces fewer errors, closely matching the GT with minimal visible inconsistencies. In addition, compared with other methods, our method achieves the best results for all four metrics, demonstrating its excellent performance.

3) Results on Pavia Center Dataset: Fig. 7 illustrates the fusion results of each method, while Table III presents

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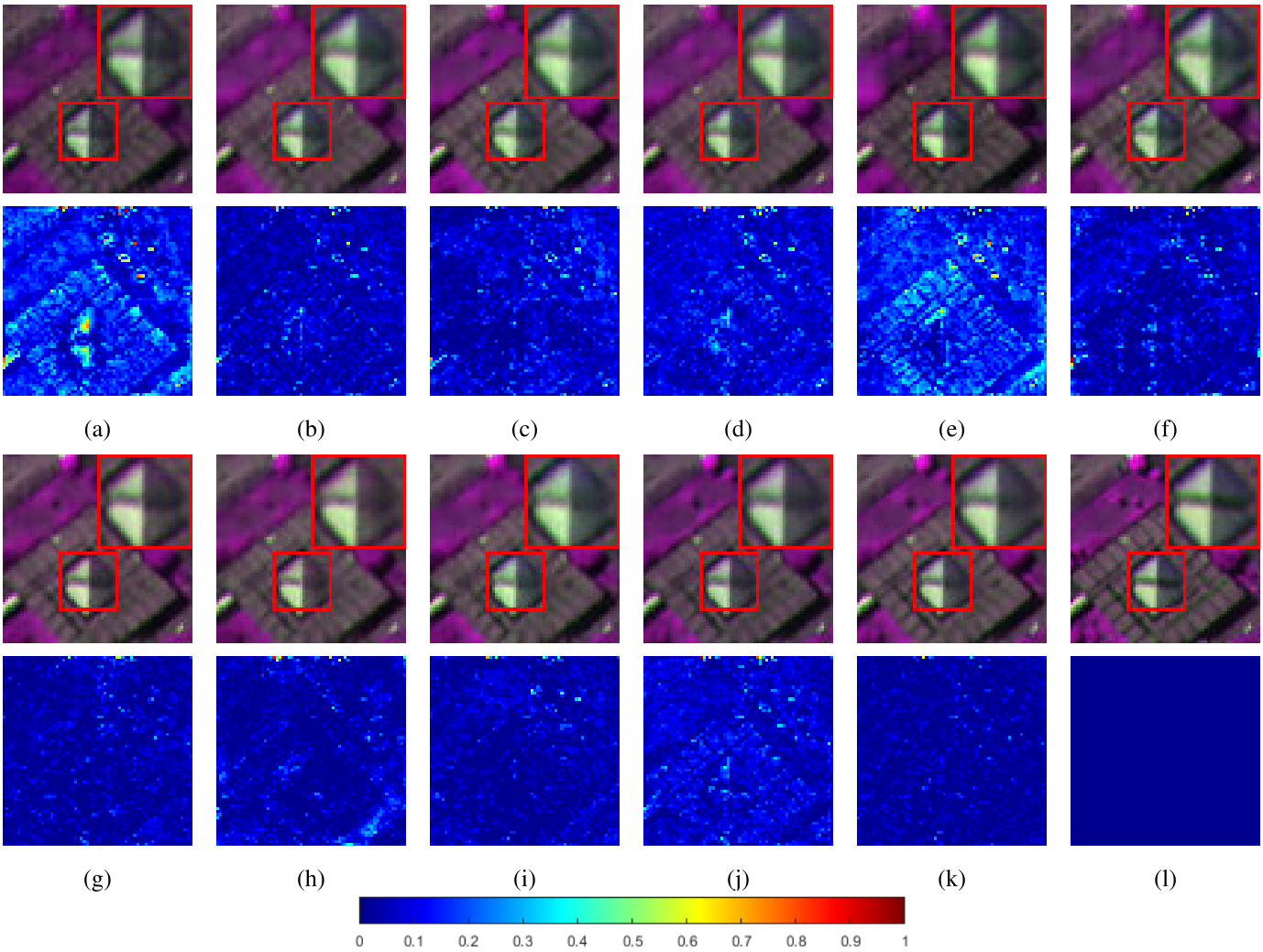


Fig. 7. Fusion results of different fusion methods on the Pavia Center dataset. The first row displays the pseudo-color RGB images of the fusion results, while the second row presents the corresponding absolute error maps. (a) SSR-NET, (b) Fusformer, (c) PSRT, (d) MCT-Net, (e) QIS-GAN, (f) CSSNet, (g) AMGSGAN, (h) SDAGE, (i) NGST_Net, (j) CYformer, (k) proposed, and (l) GT.

TABLE II
EXPERIMENTAL RESULTS ON THE HOUSTON DATASET, WITH THE BEST VALUES MARKED IN BOLD

Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓	Paras(M)
SSR-NET(2021)	42.385	0.714	5.289	9.302	0.560
Fusformer(2022)	47.953	0.847	3.982	4.907	0.166
PSRT(2023)	45.875	0.798	3.674	7.384	0.313
MCT-Net(2023)	49.630	0.867	3.141	4.155	6.934
QIS-GAN(2023)	45.835	0.824	4.269	8.406	0.378
CSSNet(2023)	49.593	0.869	5.230	4.516	2.244
AMGSGAN(2023)	48.430	0.853	3.495	4.955	4.065
SDAGE(2024)	46.430	0.798	4.367	5.680	6.754
NGST_Net(2024)	47.810	0.856	3.631	7.377	0.222
CYformer(2025)	47.171	0.846	3.493	6.143	0.874
Proposed	**50.475**	**0.892**	**2.961**	**3.704**	2.123

the evaluation metrics for each fusion method on the Pavia Center dataset. As shown in Fig. 7, SSR-NET, SDAGE, and NGST_Net exhibit noticeable spectral distortions in various colors, particularly with significant deviations in the purple areas. On the other hand, Fusformer, CSSNet, PSRT, MCT-Net, QIS-GAN, CYformer, and AMGSGAN display varying degrees of spatial distortion. Although some of them maintain better color retention, the image details become blurred, especially along the edges and textures, failing to capture the spatial information present in the original image. In comparison, CSGAV preserves better spectral and spatial information while exhibiting the smallest error and achieving the best performance across four metrics, demonstrating its significant advantage in fusing spectral and spatial information.

To further highlight the superior spectral preservation of CSGAV, we next compare its performance with that of SOTA methods in terms of spectral correlation on three datasets, as depicted in Figs. 8–10, the spectral curves of the fused images are presented to illustrate potential spectral distortions. The pixel is randomly selected from the test image to compare the spectral similarity between the GT and the fused image. To improve clarity, we focused on a subset of channels, specifically channels 30–45. Spectral curves can reflect the values of different spectral bands in HR-HSI. It can be seen that the fused images obtained by the CSGAV are consistently

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TABLE III
EXPERIMENTAL RESULTS ON THE PAVIA CENTER DATASET, WITH THE BEST VALUES MARKED IN BOLD

Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓	Paras(M)
SSR-NET(2021)	33.921	0.939	6.923	4.532	0.281
Fusformer(2022)	33.307	0.919	5.824	4.987	0.145
PSRT(2023)	31.439	0.899	5.711	5.473	0.289
MCT-Net(2023)	35.040	0.944	4.850	3.990	4.016
QIS-GAN(2023)	32.612	0.921	5.826	5.247	0.325
CSSNet(2023)	34.161	0.943	7.371	4.043	1.161
AMGSGAN(2023)	36.704	0.963	5.011	2.899	3.963
SDAGE(2024)	28.957	0.844	6.332	6.878	6.456
NGST_Net(2024)	35.930	0.953	5.211	3.357	0.202
CYformer(2025)	38.838	0.965	4.937	2.673	0.781
Proposed	**39.930**	**0.971**	**4.592**	**2.217**	1.708

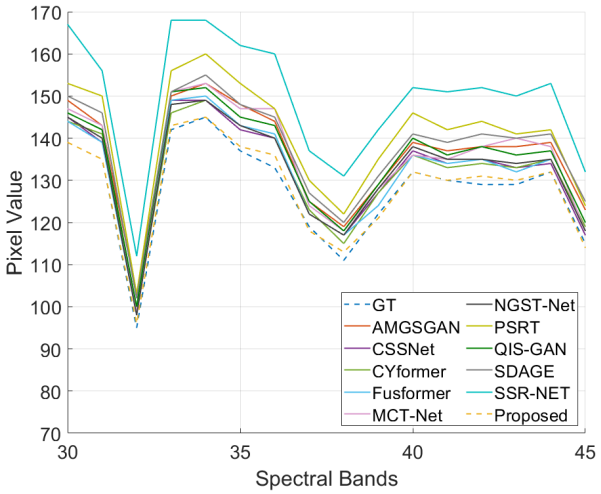


Fig. 8. Spectral curves of the proposed method and the SOTA MHIF methods on the Botswana dataset.

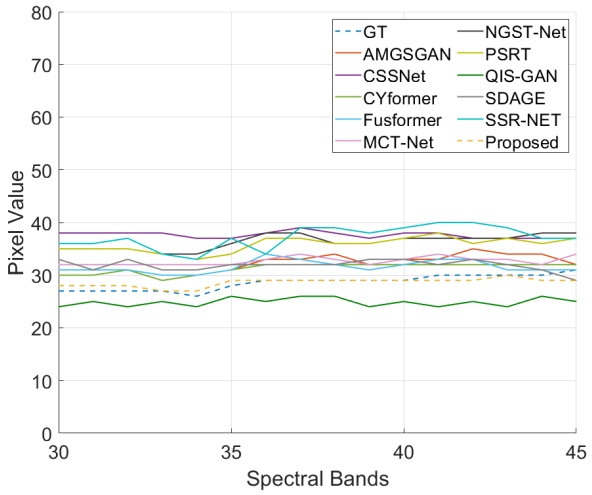


Fig. 9. Spectral curves of the proposed method and the SOTA MHIF methods on the Houston dataset.

closest to the GT in spectral vectors under different input conditions. This is a significant indication that CSGAV can effectively mitigate spectral distortion, which is especially crucial in the MHIF task.

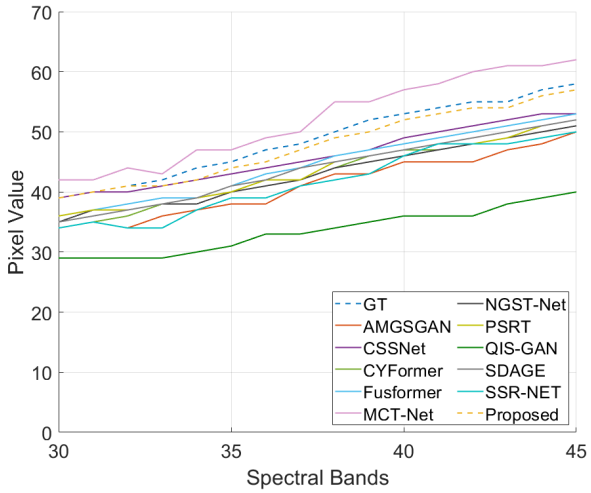


Fig. 10. Spectral curves of the proposed method and the SOTA MHIF methods on the Pavia Center dataset.

D. Experimental Results on Real Dataset

To validate the efficacy of our model on the real-world dataset, we take the WorldView-3 dataset as an example and provide visual comparisons in Fig. 11. The fused image size is $B \times 1024 \times 1024$, with B indicating the number of bands. Due to its large size, it is not feasible to directly display it in the article. To enhance visualization, we focus on a specific area and provide a close-up view. Due to the absent of a GT image for the full-resolution case, we rely on the EXP method [50] as a spectral reference [9]. Note that EXP is an upsampling method rather than a fusion method; therefore, it is excluded from the comparison.

Notably, the outcomes obtained from SSR-NET, MCT-Net, CSSNet, and SDAGE exhibit significant spectral distortion, leading to noticeable changes in the color of the red roofs. The results of Fusformer, PSRT, QIS-GAN, NGST_Net, and AMGSGAN suffer from spatial distortion, while the result of CYformer appears oversaturated. In comparison, our method produces a more natural result and maintains better spectral fidelity, as shown in the comparison with the EXP reference.

To further quantify the fusion quality, we provide objective evaluation metrics in Table IV. It can be observed that our method achieves the highest performance in terms of $D_s \downarrow$ and QNR↑ metrics with a clear advantage, while also maintaining competitive performance in the $D_\lambda \downarrow$ metric.

E. Comparison With Upsampling Methods

To evaluate the effectiveness of our proposed SGA module, we conducted comparison experiments against various upsampling methods, including bicubic, bilinear, nearest, and TC. Figs. 12–14 present the pseudo-color RGB images of the fusion results along with their corresponding absolute error maps between the fusion results and the reference images. The detailed evaluation metrics, including PSNR↑, UIQI↑, SAM↓, and ERGAS↓, across three datasets are provided in Tables V–VII. Based on the experimental data and the fused images, it is clear that all performance metrics of the SGA

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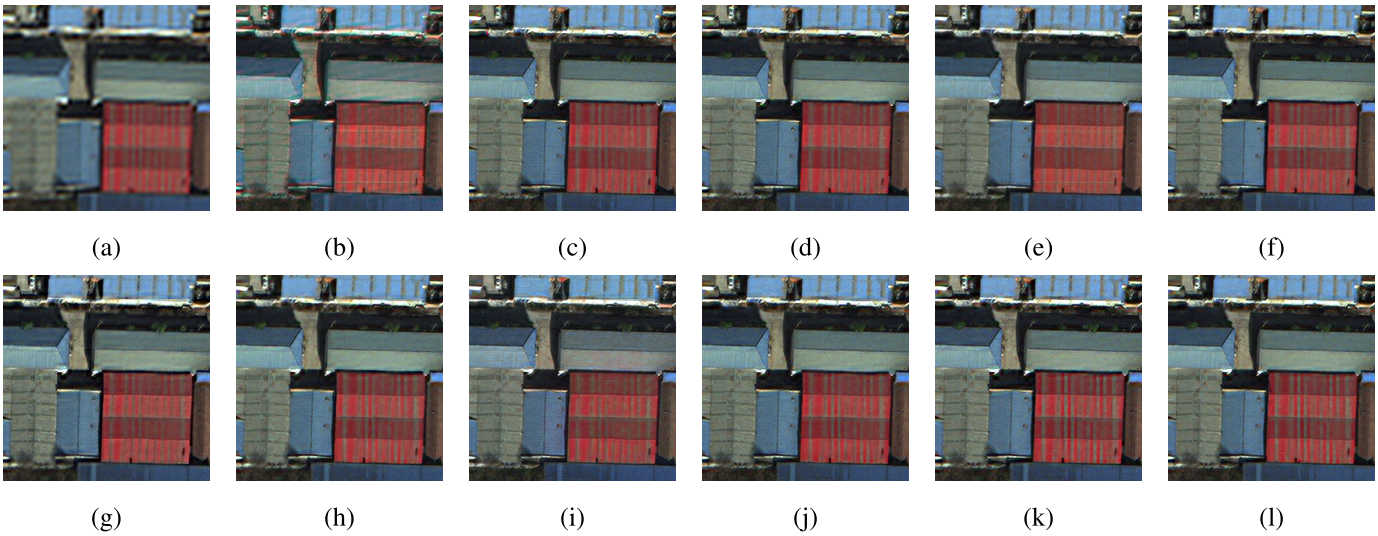


Fig. 11. Fusion results of different fusion methods on the WorldView-3 dataset. The first row displays the pseudo-color RGB images of the fusion results. (a) EXP, (b) SSR-NET, (c) Fusformer, (d) PSRT, (e) MCT-Net, (f) QIS-GAN, (g) CSSNet, (h) AMGSGAN, (i) SDAGE, (j) NGST_Net, (k) CYformer, and (l) proposed.

TABLE IV

QUANTITATIVE ASSESSMENT OF REAL EXPERIMENTS ON THE WORLDVIEW-3 DATASET, WITH THE BEST VALUES MARKED IN BOLD

Method	$D_\lambda \downarrow$	$D_s \downarrow$	QNR ↑
SSR-NET(2021)	0.0390	0.0710	0.8931
Fusformer(2022)	**0.0187**	0.0529	0.9299
PSRT(2023)	0.0209	0.0561	0.9245
MCT-Net(2023)	0.0504	0.0777	0.8763
QIS-GAN(2023)	0.0333	0.0687	0.9009
CSSNet(2023)	0.0727	0.0777	0.8560
AMGSGAN(2023)	0.0232	0.0582	0.9204
SDAGE(2024)	0.0568	0.0663	0.8814
NGST_Net(2024)	0.0244	0.0600	0.9177
CYformer(2025)	0.0239	0.0583	0.9197
Proposed	0.0231	**0.0467**	**0.9313**

TABLE V

USING DIFFERENT UPSAMPLING METHODS IN BOTSWANA DATASET, WITH THE BEST VALUES MARKED IN BOLD

Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓	Paras(M)
Bicubic	38.592	0.973	1.680	0.437	1.628
Bilinear	38.419	0.973	1.736	0.443	1.628
Nearest	38.871	0.974	1.682	0.421	1.628
TC	38.777	0.973	1.698	0.428	2.974
Proposed	**42.554**	**0.986**	**1.600**	**0.282**	2.134

module have shown improvement compared to the simple upsampling method. Furthermore, the residual maps demonstrate that the SGA module effectively minimizes errors. These results fully demonstrate that SGA can effectively utilize the spatial information of MSI to guide the upsampling of HSI, exhibiting satisfactory robustness and accuracy.

F. Ablation Study

To assess the contribution of each component in the proposed CSGAV model, we performed a series of ablation experiments on the three simulated datasets mentioned above.

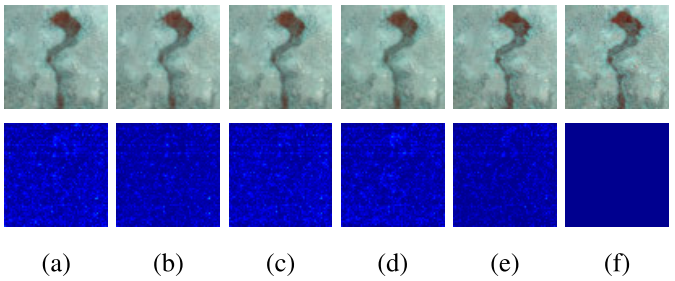


Fig. 12. Fusion results of various upsampling methods on the Botswana dataset. The first row displays the pseudo-color RGB images of the fusion outcomes, while the second row presents the corresponding absolute error maps. (a) Bicubic, (b) bilinear, (c) nearest, (d) TC, (e) proposed, and (f) GT.

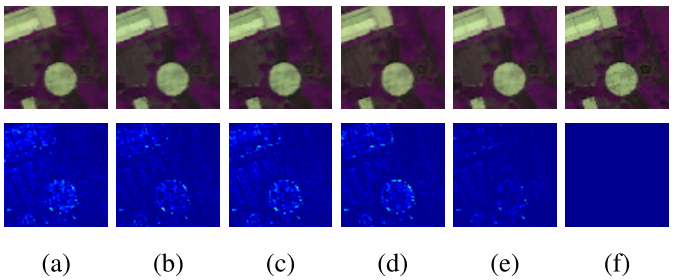


Fig. 13. Fusion results of various upsampling methods on the Houston dataset. The first row displays the pseudo-color RGB images of the fusion outcomes, while the second row presents the corresponding absolute error maps. (a) Bicubic, (b) bilinear, (c) nearest, (d) TC, (e) proposed, and (f) GT.

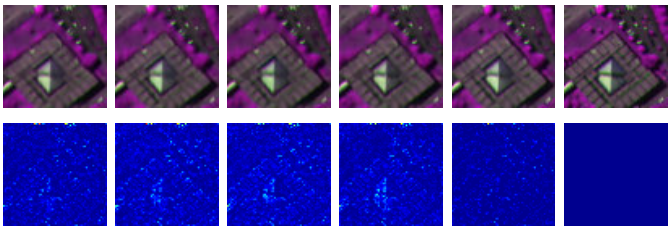
TABLE VI

USING DIFFERENT UPSAMPLING METHODS IN HOUSTON DATASET, WITH THE BEST VALUES MARKED IN BOLD

Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓	Paras(M)
Bicubic	48.355	0.864	3.144	4.680	1.624
Bilinear	48.260	0.867	3.252	5.587	1.624
Nearest	46.848	0.858	3.212	5.311	1.624
TC	50.063	0.880	3.186	3.906	2.951
Proposed	**50.475**	**0.892**	**2.961**	**3.704**	2.123

Specifically, we trained and tested the models in four configurations: without SGA or Transformer or VAE, without

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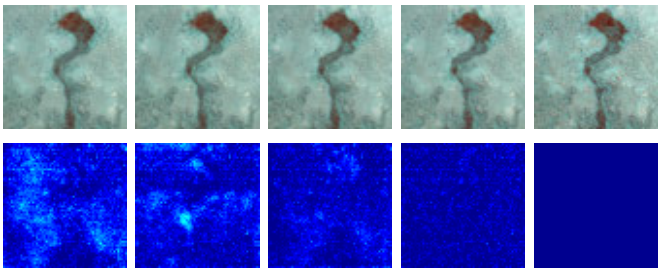
(a) (b) (c) (d) (e) (f)

Fig. 14. Fusion results of various upsampling methods on the Pavia Center dataset. The first row displays the pseudo-color RGB images of the fusion outcomes, while the second row presents the corresponding absolute error maps. (a) Bicubic, (b) bilinear, (c) nearest, (d) TC, (e) proposed, and (f) GT.

TABLE VII

USING DIFFERENT UPSAMPLING METHODS IN PAVIA CENTER DATASET, WITH THE BEST VALUES MARKED IN BOLD

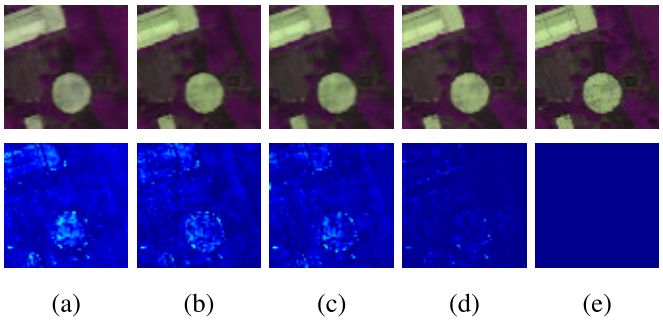
Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓	Paras(M)
Bicubic	38.730	0.967	4.720	2.502	1.458
Bilinear	38.648	0.968	4.723	2.536	1.458
Nearest	38.493	0.968	4.789	2.568	1.458
TC	38.192	0.966	4.730	2.651	2.124
Proposed	**39.930**	**0.971**	**4.592**	**2.217**	1.708



(a) (b) (c) (d) (e)

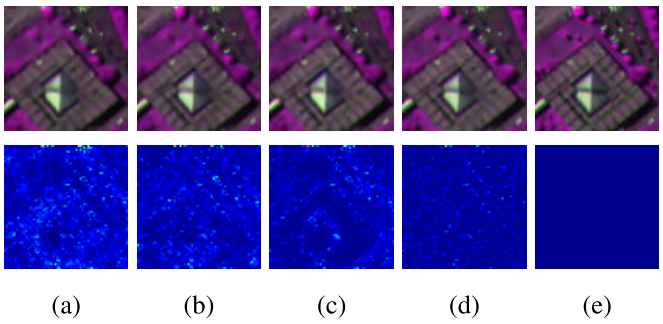
Fig. 15. Visualization results of ablation study on Botswana dataset. The first row shows pseudo-color RGB images, and the second row shows the corresponding error maps. (a) W/o SGA or Transformer or VAE, (b) w/o Transformer or VAE, (c) w/o VAE, (d) proposed, and (e) GT.

Transformer or VAE, without VAE, and using the full CSGAV model. Note that, in the absence of SGA, we use bicubic interpolation as the upsampling method for our model. Figs. 15–17 display the visual results of the fusion results for each method, while Table VIII presents the evaluation metrics for each fusion method across the three datasets. The impact of each component on performance was assessed across the Botswana, Houston, and Pavia Center datasets. These results demonstrate clear visual and quantitative improvements across three configurations, highlighting the progressive enhancement in both spectral and spatial information as different modules are introduced. Each module—including SGA, Transformer, and VAE—contributes to these improvements. Additionally, the residual error maps show a consistent reduction in errors across the configurations, with the discrepancies between the reconstructed and GT images becoming smaller with the addition of our proposed modules. This confirms the cumulative effectiveness of each module in refining image quality and reducing reconstruction errors.



(a) (b) (c) (d) (e)

Fig. 16. Visualization results of ablation study on Houston dataset. The first row shows pseudo-color RGB images, and the second row shows the corresponding error maps. (a) W/o SGA or Transformer or VAE, (b) w/o Transformer or VAE, (c) w/o VAE, (d) proposed, and (e) GT.



(a) (b) (c) (d) (e)

Fig. 17. Visualization results of ablation study on Pavia Center dataset. The first row shows pseudo-color RGB images, and the second row shows the corresponding error maps. (a) W/o SGA or Transformer or VAE, (b) w/o Transformer or VAE, (c) w/o VAE, (d) proposed, and (e) GT.

In addition to visual improvements, key performance metrics such as PSNR↑ and ERGAS↓ also show significant enhancements across the datasets. For instance, on the Botswana dataset, after adding SGA and VAET, the PSNR↑ increased by 22.7% (from 34.650 to 42.554), and the ERGAS↓ decreased by 61.7% (from 0.738 to 0.282), demonstrating the powerful impact of SGA and VAET on improving overall image quality. Similar trends were observed in the Houston and Pavia Center datasets, where the inclusion of each module led to steady improvements in both PSNR↑ and ERGAS↓. On the Houston dataset, the PSNR↑ increased by 14.9% (from 43.903 to 50.475), and the ERGAS↓ decreased by 67.0% (from 11.259 to 3.704). The Pavia Center dataset showed a PSNR↑ improvement of 24.7% (from 32.023 to 39.930) and a decrease in ERGAS↓ by 56.5% (from 5.106 to 2.217).

Overall, these results demonstrate that each component of our module, including the SGA, Transformer, and VAE, plays a crucial role in enhancing performance.

G. Parameter Analysis

In the SGA module, we select m most similar MSI nodes for each HSI node to guide the construction of the graph structure. To investigate the impact of varying m values on the fusion results, we conducted experiments on the Botswana dataset. As shown in Table IX, smaller values of m (e.g., 1–4) lead to inferior performance across all four metrics (PSNR, UIQI, SAM, ERGAS), indicating insufficient information extracted for accurate fusion. For larger values of m (e.g., 6–10), the

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TABLE VIII
EFFECTIVENESS ANALYSIS OF THE PROPOSED MODULES, WITH THE BEST VALUES MARKED IN BOLD

Database	Models	SGA	Transformer	VAE	PSNR ↑	UIQI ↑	SAM ↓	ERGAS ↓	Paras(M)
Botswana	w/o SGA or Transformer or VAE	×	×	×	34.650	0.960	1.872	0.738	1.022
	w/o Transformer or VAE	✓	×	×	36.083	0.968	1.730	0.622	1.527
	w/o VAE	✓	✓	×	36.802	0.973	1.695	0.570	1.685
	Proposed	✓	✓	✓	**42.554**	**0.986**	**1.600**	**0.282**	2.134
Houston	w/o SGA or Transformer or VAE	×	×	×	43.903	0.761	4.516	11.259	1.018
	w/o Transformer or VAE	✓	×	×	46.925	0.836	3.739	5.513	1.516
	w/o VAE	✓	✓	×	47.590	0.834	3.882	5.665	1.673
	Proposed	✓	✓	✓	**50.475**	**0.892**	**2.961**	**3.704**	2.123
Pavia Centre	w/o SGA or Transformer or VAE	×	×	×	32.023	0.915	5.712	5.106	0.852
	w/o Transformer or VAE	✓	×	×	34.814	0.951	5.191	3.738	1.102
	w/o VAE	✓	✓	×	36.272	0.958	5.081	3.210	1.259
	Proposed	✓	✓	✓	**39.930**	**0.971**	**4.592**	**2.217**	1.708

TABLE IX
ANALYSIS OF DIFFERENT VALUES OF m ON BOTSWANA DATASET, WITH THE BEST RESULTS MARKED IN BOLD

m	PSNR↑	UIQI↑	SAM↓	ERGAS↓
1	41.562	0.986	1.608	0.327
2	42.308	0.986	1.627	0.284
3	40.547	0.980	1.622	0.342
4	42.478	0.986	1.622	0.283
5	42.554	**0.986**	**1.600**	**0.282**
6	42.021	0.985	1.607	0.311
7	40.807	0.984	1.609	0.305
8	**42.604**	0.986	1.621	0.283
9	42.362	0.986	1.624	0.285
10	42.358	0.986	1.626	0.309

TABLE X
EVALUATION RESULTS ON DIFFERENT TRAINING AND TESTING DATASETS WITH AND WITHOUT NOISE, WITH THE BEST VALUES MARKED IN BOLD

Dataset	Method	PSNR↑	UIQI↑	SAM↓	ERGAS↓
Botswana-2	w/o noise	39.085	0.982	1.702	0.434
	Proposed	**39.638**	**0.983**	**1.692**	**0.411**
Botswana-3	w/o noise	36.680	0.980	1.820	0.592
	Proposed	**37.225**	**0.981**	**1.809**	**0.562**
Houston-2	w/o noise	50.348	0.888	3.092	3.999
	Proposed	**50.405**	**0.891**	**3.082**	**3.774**
Houston-3	w/o noise	50.070	0.884	3.300	4.155
	Proposed	**50.188**	**0.888**	**3.296**	**3.884**
Pavia Centre-2	w/o noise	39.457	0.970	4.862	2.295
	add noise	**39.623**	**0.970**	**4.785**	**2.276**
Pavia Centre-3	w/o noise	38.981	0.969	5.264	2.389
	Proposed	**39.228**	**0.969**	**5.168**	**2.388**

overall performance also degrades, suggesting that the added complexity introduces unnecessary noise. In contrast, $m = 5$ achieves the best performance across the majority of metrics. Therefore, we select $m = 5$ as the number of most similar MSI nodes for each HSI node in the SGA module. This moderate number of nodes ensures that the graph structure remains informative, balancing both spectral and spatial information without introducing unnecessary noise.

H. Performance of CSGAV in Robustness and Generalization

To validate the generalization capabilities of CSGAV, we conducted experiments on the simulated datasets which are generated through different methods. Specifically, we initially applied an 8 × 8, 4 × 4, and 2 × 2 anti-aliasing Gaussian filter with a standard deviation of 2 to the HR-HSI images on the Botswana, Houston, and Pavia Center datasets, respectively. And this operation can generate a total of nine datasets: Botswana-1 (with 8 × 8 Gaussian filter), Botswana-2 (with 4 × 4 Gaussian filter), Botswana-3 (with 2 × 2 Gaussian filter), Houston-1 (with 8 × 8 Gaussian filter), Houston-2 (with 4 × 4 Gaussian filter), Houston-3 (with 2 × 2 Gaussian filter), Pavia Center-1 (with 8 × 8 Gaussian filter), Pavia Center-2 (with 4 × 4 Gaussian filter), and Pavia Center-3 (with 2 × 2 Gaussian filter). Subsequently, to validate the effectiveness of adding noise in the VAE, we trained CSGAV without noise (marked with "w/o noise") and with noise on the Botswana-1 dataset and tested it on the Botswana-2 and Botswana-3 datasets. Similarly, we trained CSGAV without noise and with noise on the Houston-1 dataset and tested it on the Houston-2 and Houston-3 datasets. We also trained CSGAV without noise and with noise on the Pavia Center-1 dataset and tested it on the Pavia Center-2 and Pavia Center-3 datasets. The results in Table X indicate that adding noise leads to significant improvements across all metrics, demonstrating the enhanced robustness and generalization capabilities of the model. Besides, we have also compared CSGAV with other SOTA methods using the same training and testing datasets, which can be seen in Tables XI–XIII. The results indicate that CSGAV demonstrates relatively stable performance across all datasets and filtering conditions, outperforming them in terms of PSNR↑ and UIQI↑ metrics. Notably, CSGAV shows significant advantages in these metrics, suggesting that it not only excels on specific datasets but also adapts effectively to different datasets and filter sizes, exhibiting strong generalization capabilities.

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TABLE XI
GENERALIZATION PERFORMANCE OF CSGAV TRAINED ON BOTSWANA-1 WHILE TESTING ON BOTSWANA-2 AND BOTSWANA-3, WITH THE BEST VALUES MARKED IN BOLD

	Method	SSR-NET	Fusformer	PSRT	MCT-Net	QIS-GAN	CSSNet	AMGSGAN	SDAGE	NGST_Net	CYformer	Proposed
Botswana-2	PSNR↑	32.121	35.317	34.048	33.811	33.542	33.788	37.424	33.045	33.198	38.305	**39.638**
	UIQI↑	0.970	0.972	0.960	0.966	0.962	0.968	0.975	0.938	0.967	0.981	**0.983**
	SAM↓	2.775	1.741	1.736	1.978	1.823	1.856	1.870	2.196	1.755	1.895	**1.692**
	ERGAS↓	1.004	0.665	0.751	0.814	0.849	0.806	0.537	0.849	0.874	0.513	**0.411**
Botswana-3	PSNR↑	31.398	32.918	32.190	33.166	32.723	31.430	37.112	32.546	31.159	36.772	**37.225**
	UIQI↑	0.969	0.971	0.956	0.965	0.961	0.963	0.975	0.936	0.965	0.980	**0.981**
	SAM↓	2.781	1.802	1.778	2.133	1.837	1.941	1.919	2.224	1.832	2.081	**1.809**
	ERGAS↓	1.090	0.897	0.947	0.869	0.937	1.087	0.555	0.904	1.118	0.613	**0.562**

TABLE XII
GENERALIZATION PERFORMANCE OF CSGAV TRAINED ON HOUSTON-1 WHILE TESTING ON HOUSTON-2 AND HOUSTON-3, WITH THE BEST VALUES MARKED IN BOLD

	Method	SSR-NET	Fusformer	PSRT	MCT-Net	QIS-GAN	CSSNet	AMGSGAN	SDAGE	NGST_Net	CYformer	Proposed
Houston-2	PSNR↑	42.364	47.459	45.544	49.207	45.223	49.556	48.038	46.195	47.339	47.134	**50.405**
	UIQI↑	0.699	0.845	0.791	0.861	0.807	0.866	0.846	0.791	0.853	0.846	**0.891**
	SAM↓	5.252	4.214	3.751	3.425	4.772	6.047	3.611	4.356	3.691	3.581	**3.082**
	ERGAS↓	9.151	5.458	7.839	4.399	10.864	4.536	5.201	5.762	7.609	6.259	**3.774**
Houston-3	PSNR↑	42.258	46.721	45.192	48.833	45.266	49.089	47.719	45.968	47.021	46.998	**50.188**
	UIQI↑	0.697	0.836	0.784	0.855	0.809	0.860	0.840	0.786	0.850	0.845	**0.888**
	SAM↓	5.252	4.491	3.926	3.756	4.750	6.754	3.787	4.413	3.838	3.855	**3.296**
	ERGAS↓	9.275	6.225	8.216	4.605	10.146	4.705	5.389	5.864	7.782	6.407	**3.884**

TABLE XIII
GENERALIZATION PERFORMANCE OF CSGAV TRAINED ON PAVIA CENTER-1 WHILE TESTING ON PAVIA CENTER-2 AND PAVIA CENTER-3, WITH THE BEST VALUES MARKED IN BOLD

	Method	SSR-NET	Fusformer	PSRT	MCT-Net	QIS-GAN	CSSNet	AMGSGAN	SDAGE	NGST_Net	CYformer	Proposed
Pavia Centre-2	PSNR↑	33.218	33.137	31.340	34.422	32.008	33.804	36.441	38.863	28.899	35.971	**39.623**
	UIQI↑	0.937	0.918	0.898	0.941	0.914	0.940	0.961	0.965	0.843	0.953	**0.970**
	SAM↓	8.033	6.052	5.868	5.406	6.577	8.286	5.318	5.434	6.400	5.393	**4.785**
	ERGAS↓	4.585	5.080	5.607	4.318	5.734	4.292	2.991	2.668	6.968	3.374	**2.276**
Pavia Centre-3	PSNR↑	33.186	32.937	31.189	34.258	32.044	33.196	36.138	38.686	28.810	35.717	**39.228**
	UIQI↑	0.936	0.917	0.896	0.939	0.915	0.935	0.960	0.964	0.842	0.951	**0.969**
	SAM↓	8.327	6.405	6.168	5.865	6.686	8.931	5.787	6.225	6.553	5.687	**5.168**
	ERGAS↓	4.601	5.190	5.730	4.371	5.709	4.624	3.095	2.710	7.028	3.489	**2.388**

TABLE XIV
TRAINING AND TESTING TIMES ON DIFFERENT DEVICES

Method	RTX 3060 GPU device		RTX 3090 GPU device	
	Train (s)	Test (s)	Train (s)	Test (s)
SSR-NET	0.019	0.018	0.016	0.017
Fusformer	0.058	0.055	0.046	0.031
PSRT	0.028	0.030	0.028	0.022
MCT-Net	0.021	0.028	0.029	0.022
QIS-GAN	0.017	0.019	0.019	0.018
CSSNet	0.024	0.027	0.028	0.023
AMGSGAN	0.079	0.032	0.033	0.021
SDAGE	0.032	0.028	0.024	0.020
NGST_Net	0.026	0.022	0.024	0.019
CYformer	0.037	0.029	0.029	0.021
CSGAV	0.036	0.029	0.031	0.023

I. Runtime Analysis

To evaluate the real-world feasibility of the proposed method across different hardware configurations, we analyzed the training and inference times of the CSGAV method on various setups, as shown in Table XIV. Specifically, both training and inference times refer to the average time to process a single input image of size $64 \times 64 \times 144$. From the table, we can observe that the proposed method maintains high computational efficiency in both training and inference stages across different GPU platforms, including the RTX 3060 and RTX 3090. These results show that CSGAV not only achieves competitive runtime performance but also outperforms other SOTA methods in fusion performance, highlighting its strong potential for real-world applications.

V. CONCLUSION AND FUTURE WORK

We propose an MSI–HSI fusion network called CSGAV, which integrates an SGA module and a VAET module. The SGA module employs an SGA mechanism to emphasize important details and mitigate modal differences, leading to

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more accurate upsampling results for HSI. Meanwhile, the VAET module improves the model's generalization and robustness by utilizing a VAE-weighted Transformer. Experimental results demonstrate that CSGAV significantly enhances performance across various datasets, confirming the effectiveness of the proposed methods.

In the future, we aim to construct more innovative networks into MHIF to further enhance performance and address emerging challenges. Furthermore, we plan to enhance model efficiency by judiciously reducing the number of parameters while maintaining high performance.

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